BME 105: Genetics in the Genomics Era

Lecture 9: Chromosomal Rearrangements

Instructor: Shelbi Russell, PhD

Course website on canvas.ucsc.edu

Chromosomal rearrangements, learning objectives:

I. Learn the **types** of chromosomal rearrangements

II. Understand the phenotypic **implications** of different rearrangement types

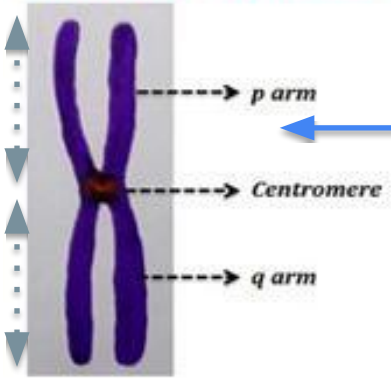
III. Understand the meiotic implications of translocations

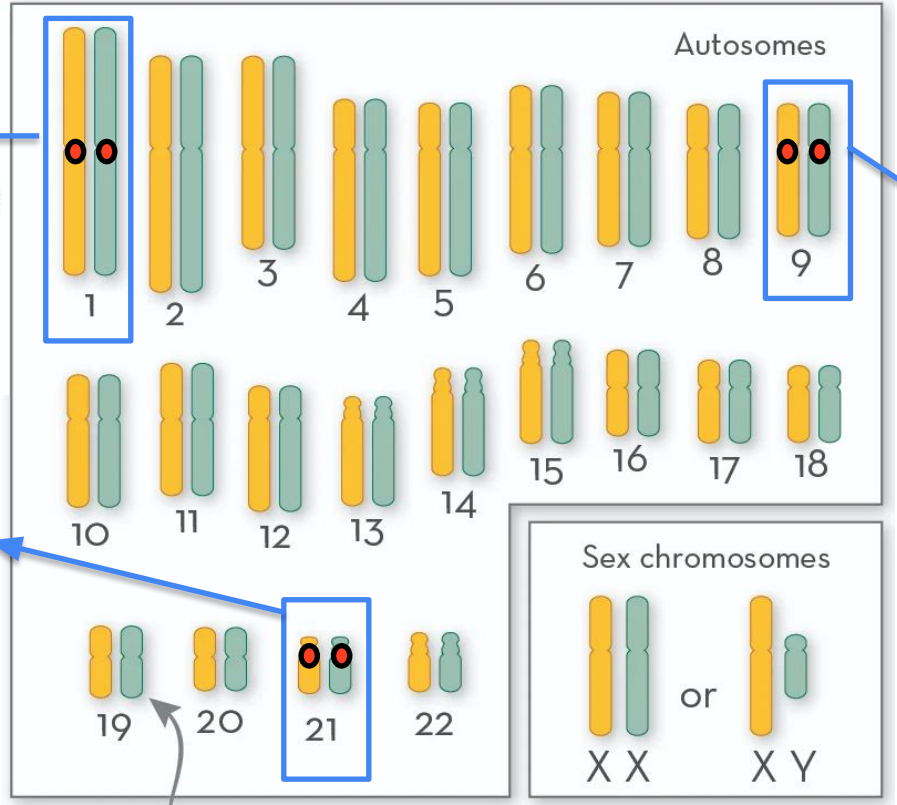
IV. Understand how rearrangements are **genotyped**

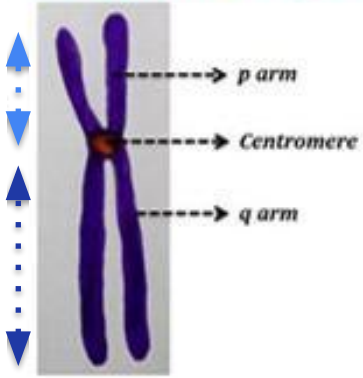
V. Learn how chromosomal rearrangements form (**mechanisms**)

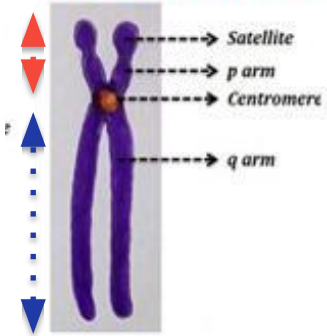
VI. Gain an **evolutionary** perspective
   - Rearrangements drive chromosomal species divergence
   - Local homology generated by transposable element insertion enables rearrangements

Types of chromosomes

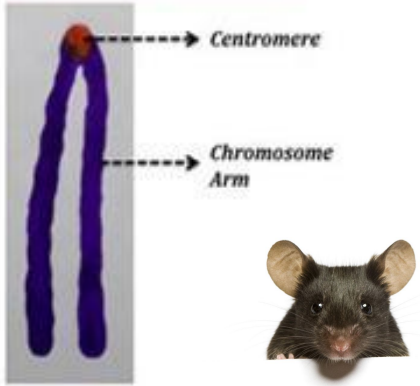






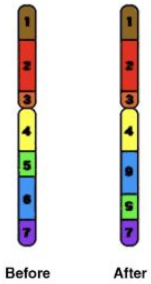


For each chromosome pair, one copy comes from each parent.

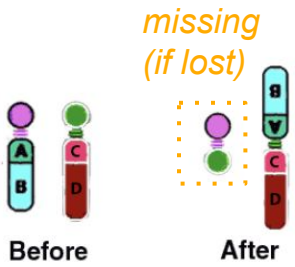


I. Summary: Types of chromosomal rearrangements

Balanced (== genetic information)

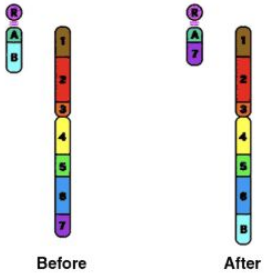


inversions



Robertsonian translocation

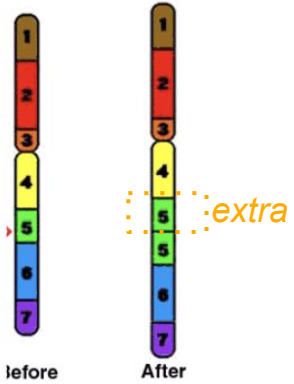
translocation



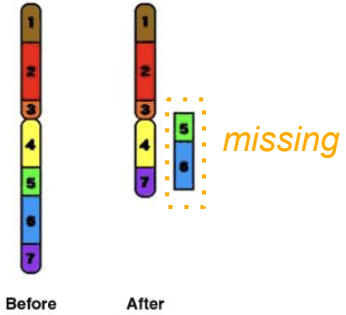
between heterologous chromosomes

Unbalanced (≠ info)

duplications



deletions



II. Phenotypic implications of chromosomal rearrangements

Balanced (== genetic information)

- No phenotype in individual
  - Unless genes are interrupted by the translocation
- Reduced fertility due to chromosomal incompatibilities in offspring

Unbalanced (≠ info)

- Phenotypic severity linked to:
  - Size of region duplicated/deleted
  - Gene content within region
- Range from mild disease to inviability

II. Phenotypic implications of chromosomal rearrangements

Human diseases caused by copy number variants

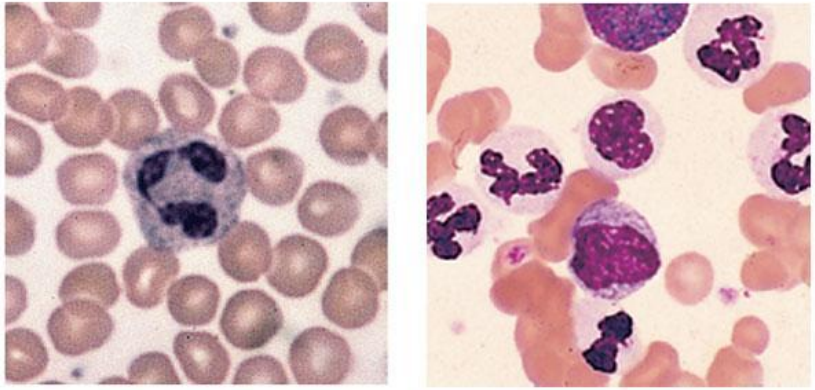
Table 2 Examples of copy number variations (CNVs) and conveyed genomic disorders[a]

Phenotype	OMIM	Locus	CNV
Mendelian (autosomal dominant)[b]			
Williams-Beuren syndrome	194050	7q11.23	del -
7q11.23 duplication syndrome	609757	7q11.23	dup +
Spinocerebellar ataxia type 20	608687	11q12	dup +
Smith-Magenis syndrome	182290	17p11.2/*RAI1*	del -
Potocki-Lupski syndrome	610883	17p11.2	dup +
HNPP	162500	17p12/*PMP22*	del -
CMT1A	118220	17p12/*PMP22*	dup +
Miller-Dieker lissencephaly syndrome	247200	17p13.3/*LIS1*	del -
Mental retardation	601545	17p13.3/*LIS1*	dup +
DGS/VCFS	188400/192430	22q11.2/*TBX1*	del -
Microduplication 22q11.2	608363	22q11.2	dup +
Adult-onset leukodystrophy	169500	*LMNB1*	dup +
Mendelian (autosomal recessive)			
Familial juvenile nephronophthisis	256100	2q13/*NPHP1*	del -
Gaucher disease	230800	1q21/*GBA*	del -
Pituitary dwarfism	262400	17q24/*GH1*	del -
Spinal muscular atrophy	253300	5q13/*SMN1*	del -
beta-thalassemia	141900	11p15/*beta-globin*	del -
alpha-thalassemia	141750	16p13.3/*HBA*	del -

Zhang et al, 2009

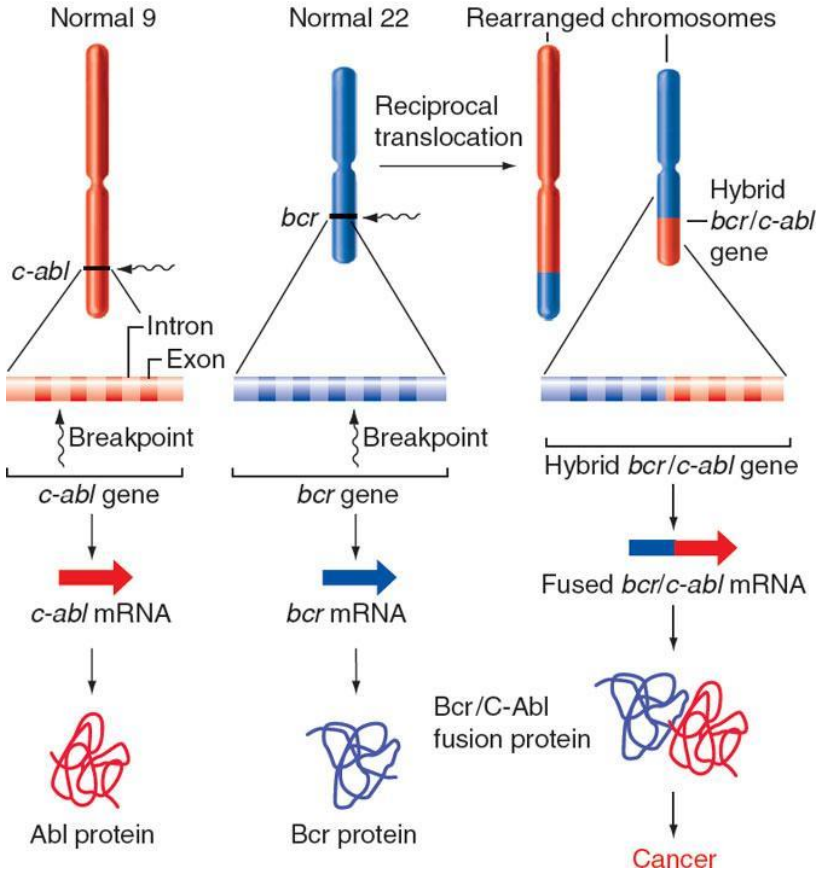
Translocations can create fusion genes

(a) Leukemia patients have too many white blood cells.



Normal Leukemic

(b) The genetic basis for chronic myelogenous leukemia



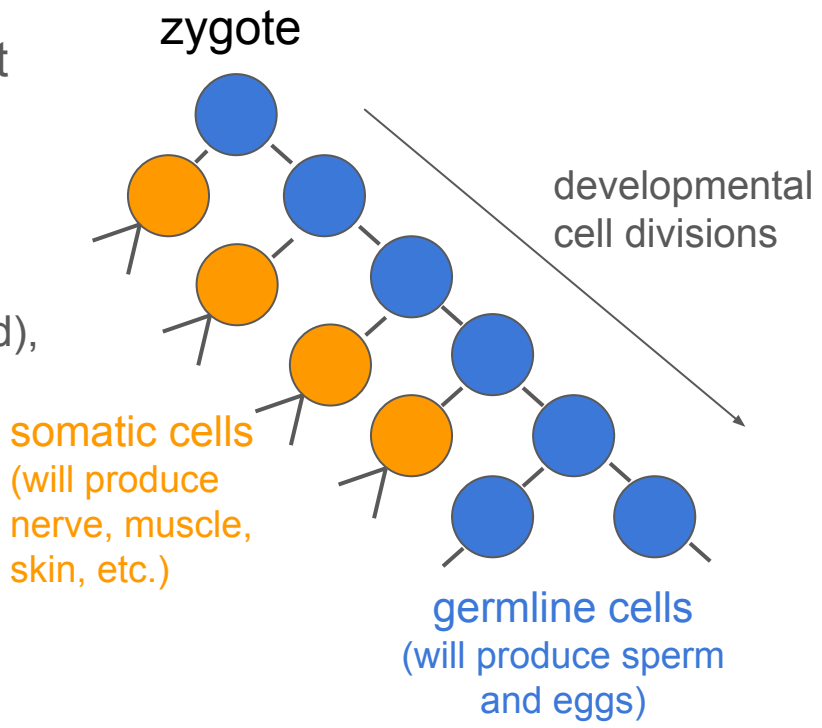
II. Phenotypic implications of chromosomal rearrangements

Timing and location of rearrangement impacts phenotype:

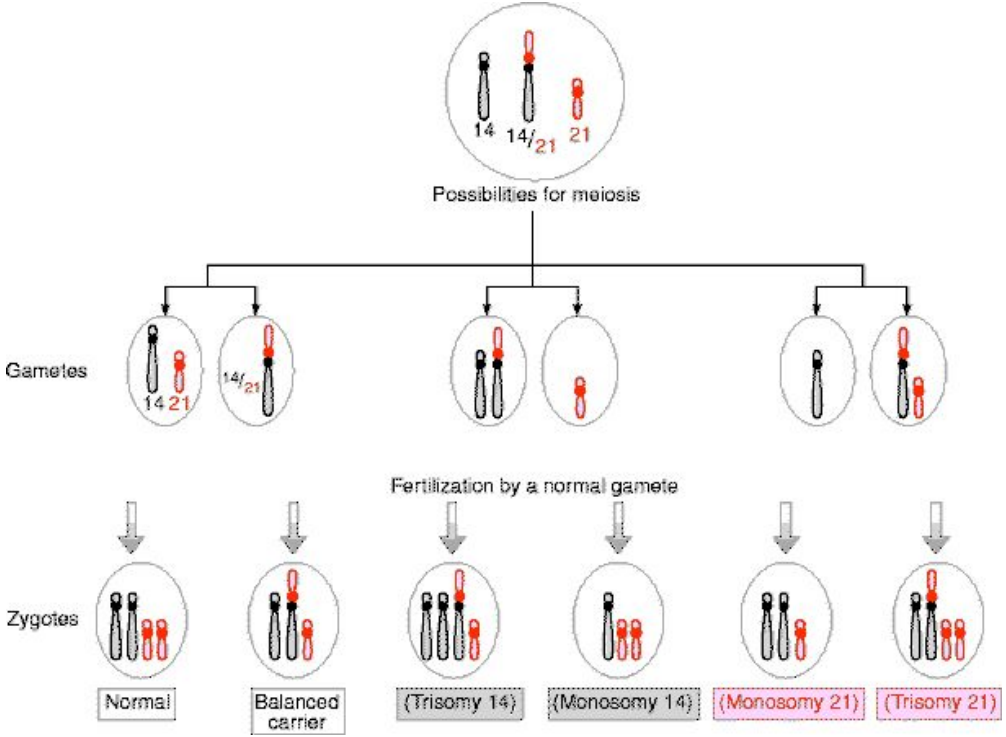
- Segregating in zygote
  → abnormal development
  → normal (*e.g.*, if balanced), **but infertile**

- *De novo* in soma
  → cancer

- *De novo* in germline
  → **infertility**



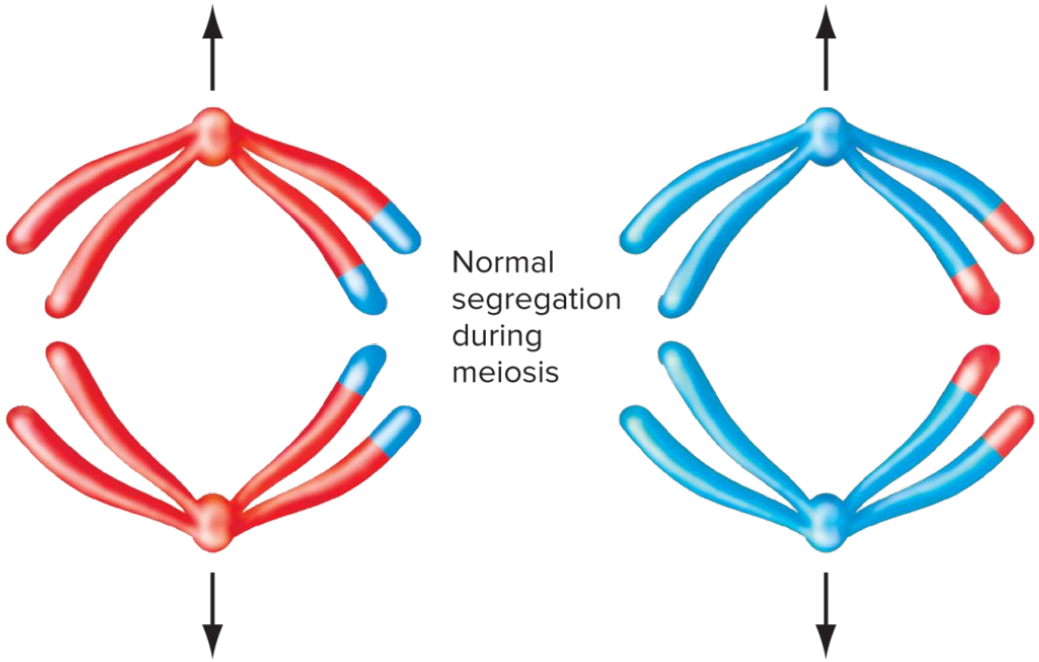
Robertsonian translocation carriers are asymptomatic, but often produce unbalanced gametes



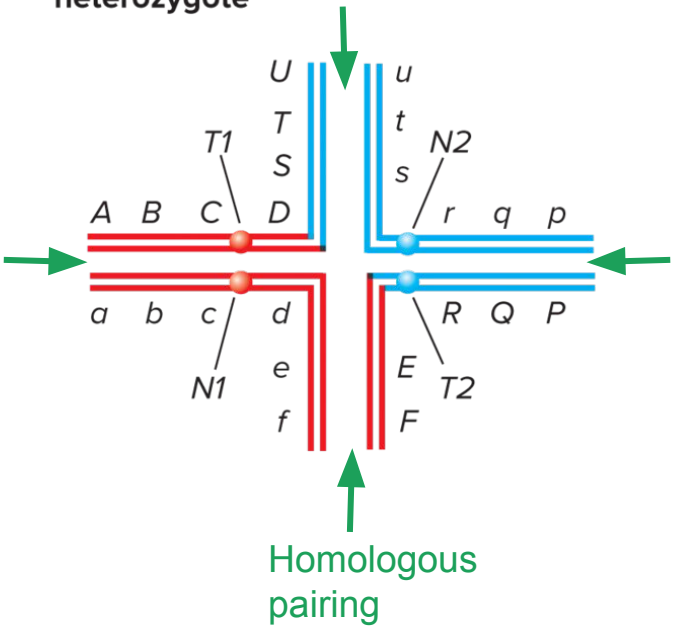
(Human Molecular Genetics, Strachan and Read 2018)

Meiotic segregation of translocation heterozygote:
Equally likely balanced/unbalanced segregation during Metaphase I

Segregation in a translocation homozygote

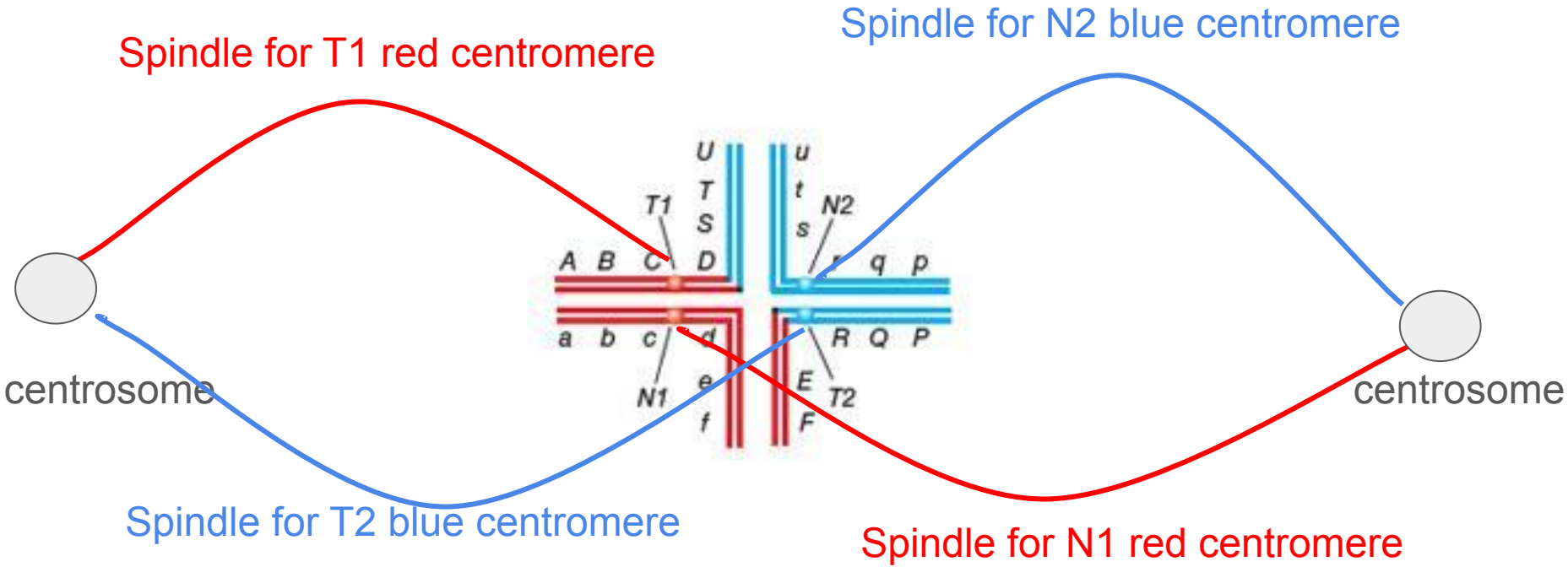


Chromosome pairing in a translocation heterozygote



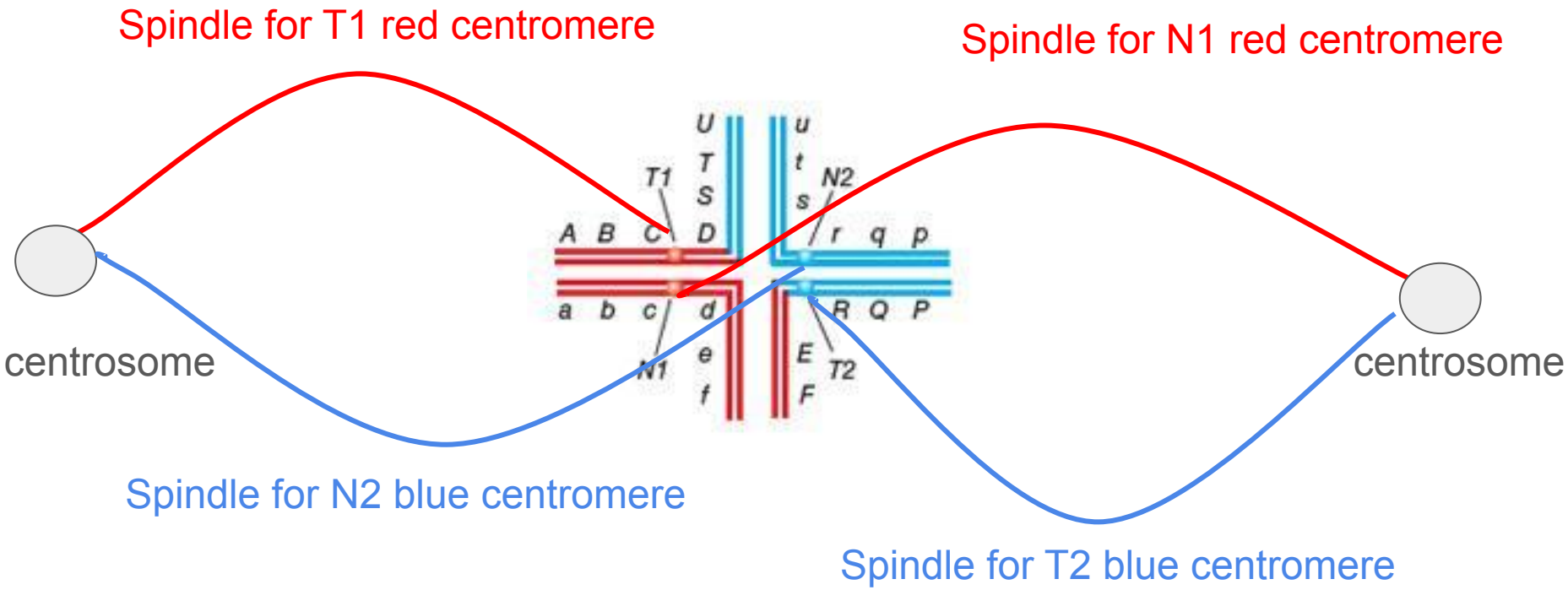
N = normal order of genes
T = translocated chromosomes

Meiotic segregation of translocation heterozygote:
Equally likely balanced/unbalanced segregation during Metaphase I



Homologous chromosomes are segregated by spindles connected to the centromere of the chromosomes

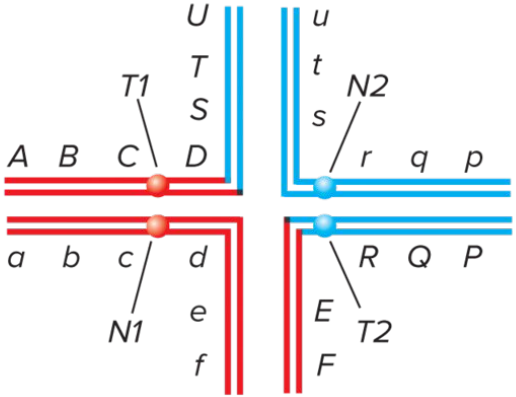
Meiotic segregation of translocation heterozygote: Centromere non-disjunction during Metaphase I is also possible



Homologous chromosomes are segregated by spindles connected to the centromere of the chromosomes

Meiotic segregation of translocation heterozygote: Centromere non-disjunction during Metaphase I is also possible

Chromosome pairing in a translocation heterozygote during Meiosis I

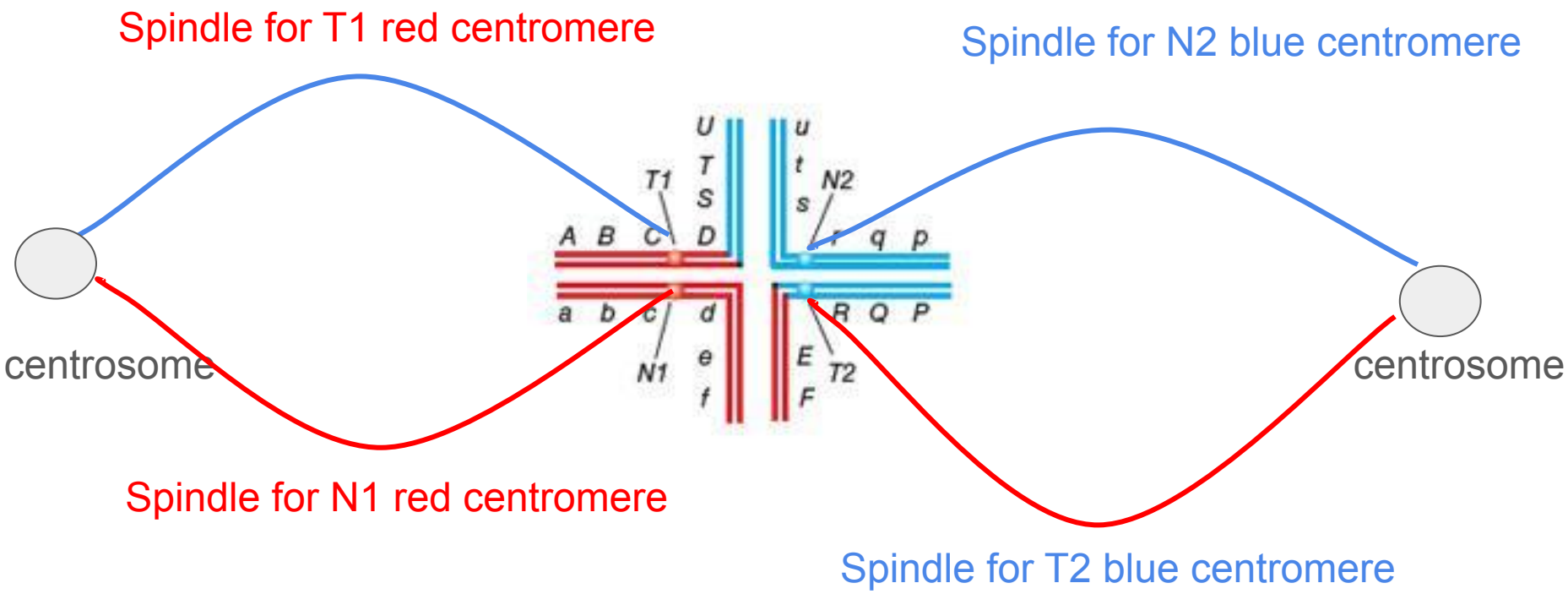


Why is Adjacent-2 less frequent?

Segregation in a translocation heterozygote

Segregation pattern	Alternate		Adjacent-1		Adjacent-2 (less frequent)	
Gametes	Balanced *N1 + N2*	Balanced *T1 + T2*	Unbalanced *T1 + N2*	Unbalanced *N1 + T2*	Unbalanced *N1 + T1*	Unbalanced *N2 + T2*
	a b c d e f / *p q r s t u*	*A B C D S T U* / *P Q R E F*	*A B C D S T U* / *p q r s t u*	*a b c d e f* / *P Q R E F*	*a b c d e f* / *A B C D S T U*	*p q r s t u* / *P Q R E F*
Type of progeny when mated with normal *abcdefpqrstu* homozygote	*abcdef pqrstu*	*ABCDEF PQRSTU*	None surviving	None surviving	None surviving	None surviving

Adjacent 2 Pattern - Unusual cruciform pairing causes rare homolog non-disjunction during Metaphase I



Homologous chromosomes are segregated by spindles connected to the centromere of the chromosomes

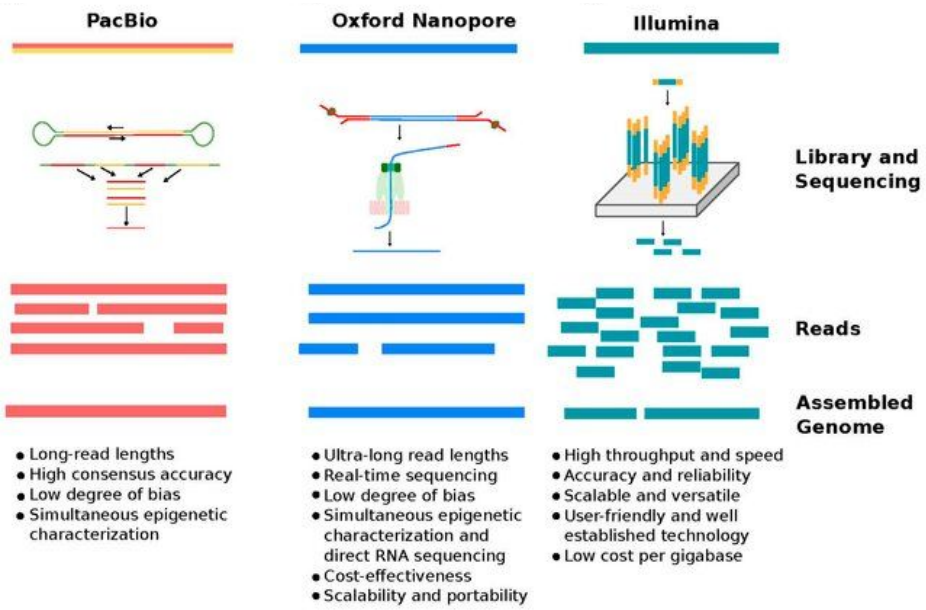
Chromosomal rearrangements in normal individuals

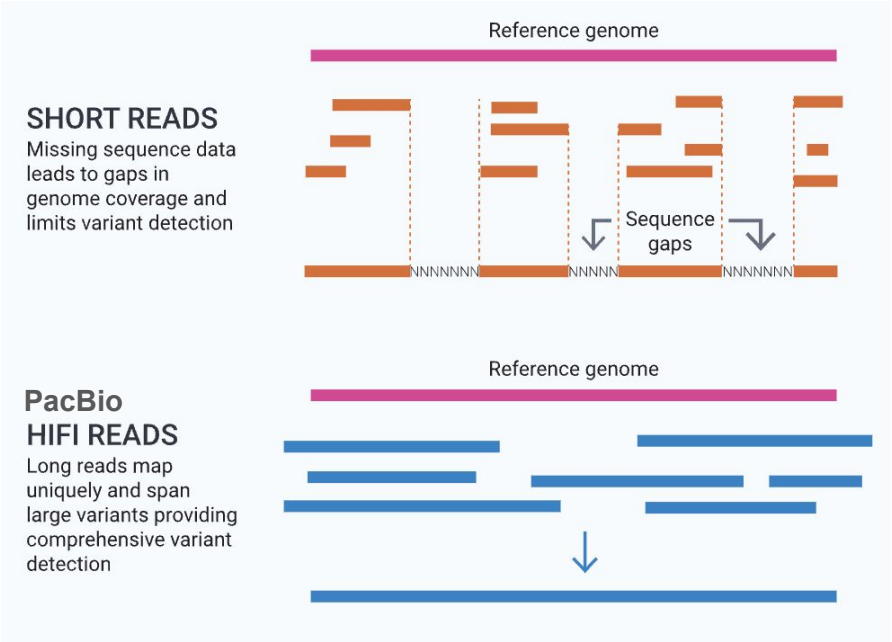
- 4.8–9.7% of the human genome contains copy number variation (CNVs)
- More than 100 genes have been observed to be completely deleted without an observable consequence
- 1 in 500 children have reciprocal translocations
- 1 in 50 children have inversions (small and large)

1000 genomes project
Copy number variation map

III. Genotyping chromosomal rearrangements

Reminder: short vs long read high throughput sequencing methods (Lecture 5)

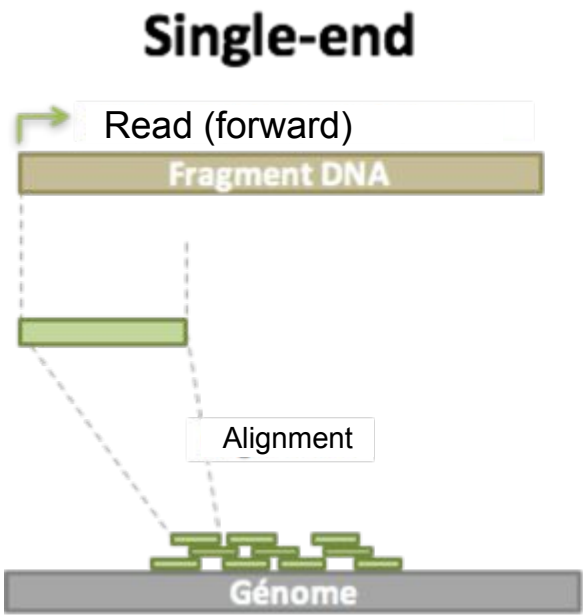


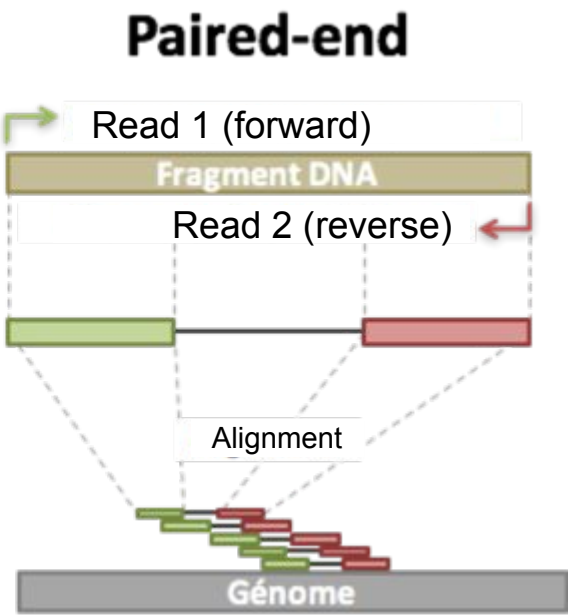


Ermini and Driguez 2024

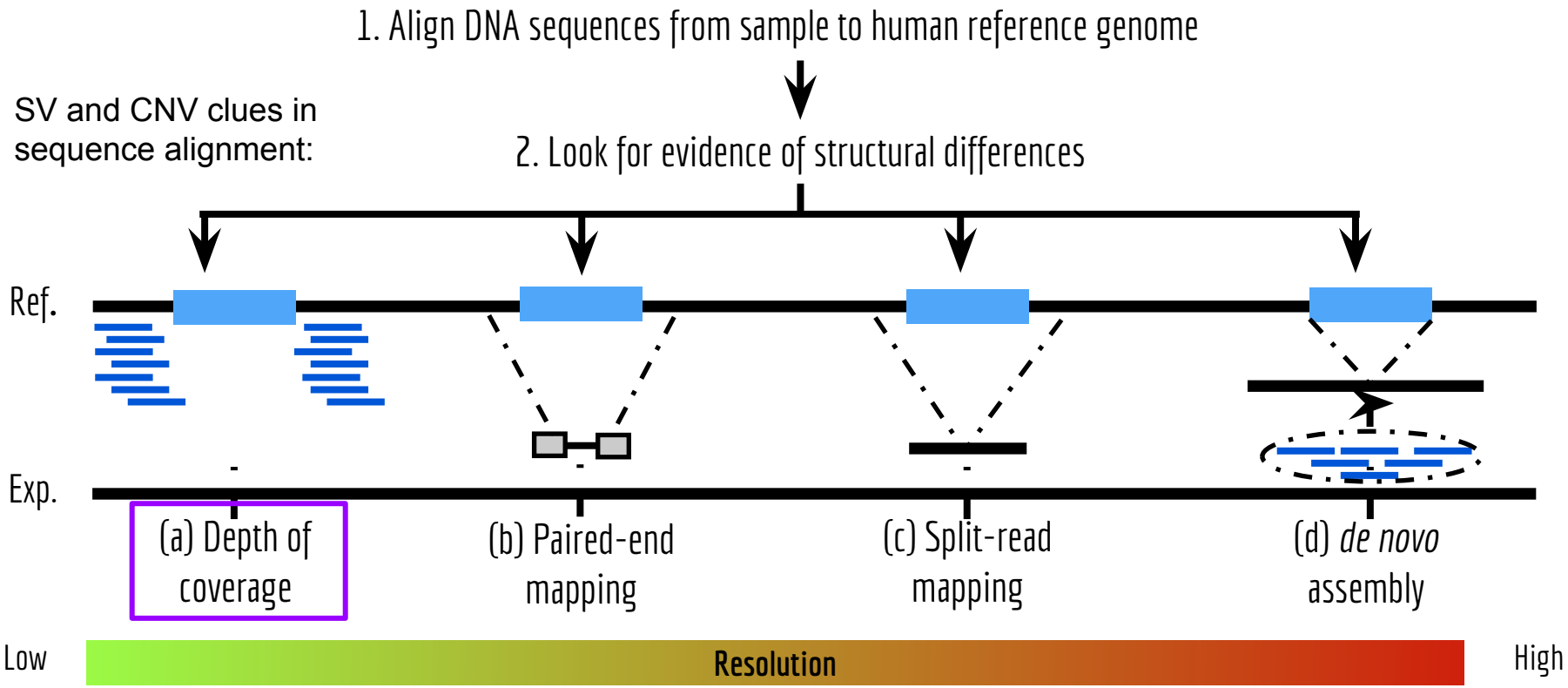
III. Genotyping chromosomal rearrangements

1. Align DNA sequences from sample to human reference genome

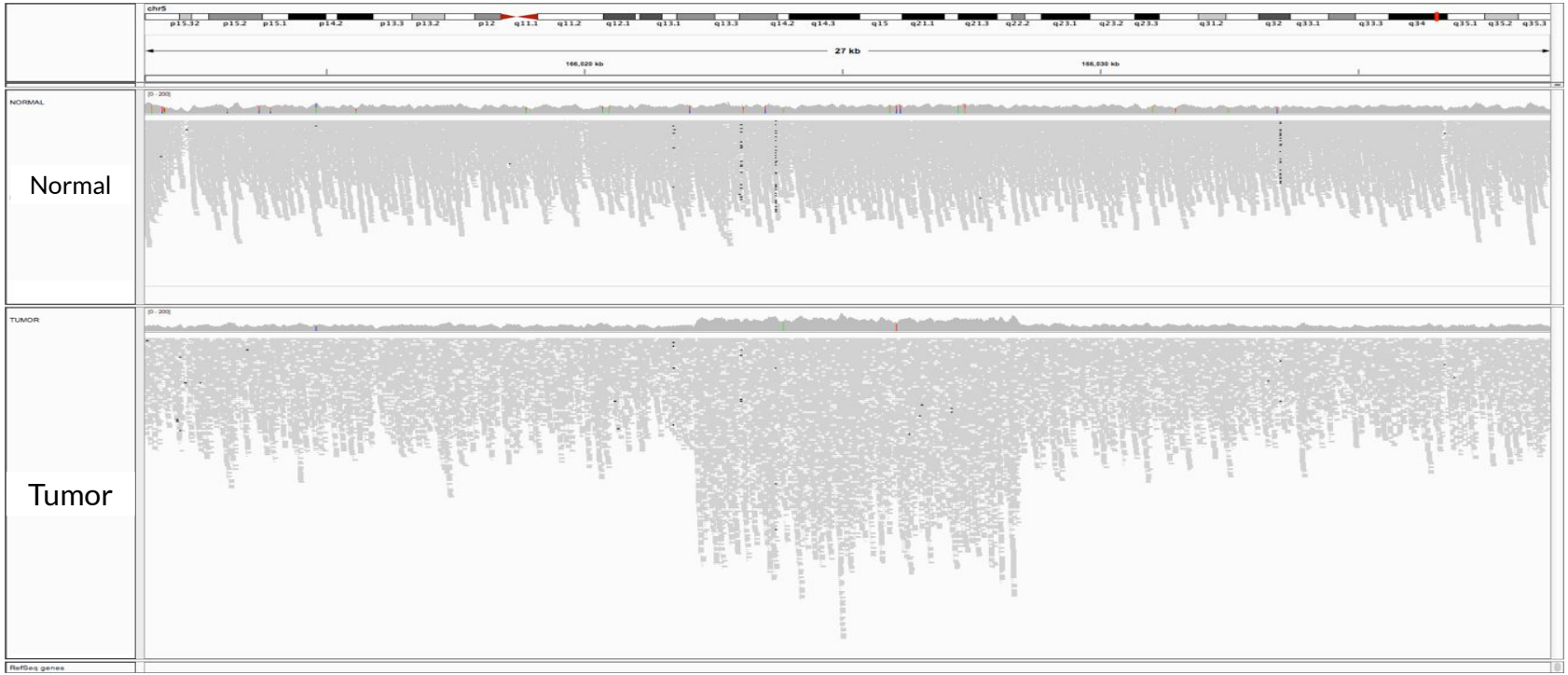




III. Genotyping chromosomal rearrangements



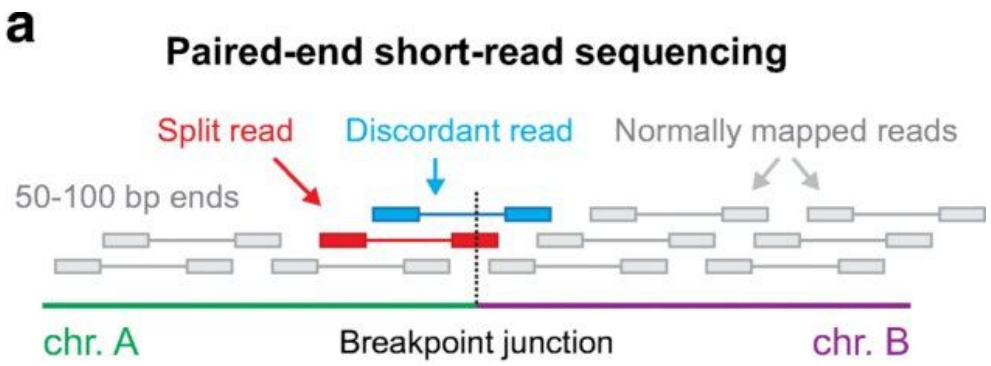
a) Copy number changes affect the depth of sequence coverage



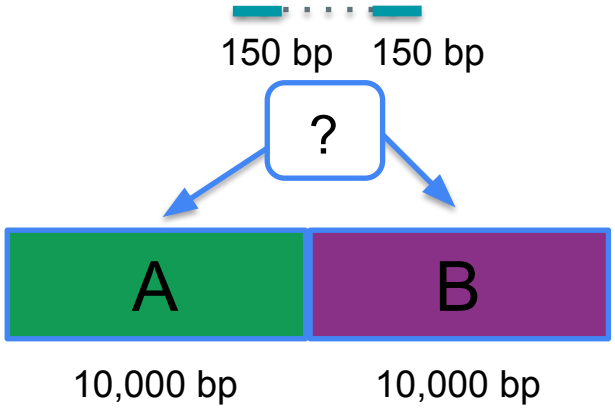
Challenges:

- need high coverage for high resolution
- deletions easier to detect than duplications
- prone to artifacts owing to repeats, GC content, etc.

b,c) Reads spanning breakpoints inform on junction and full haplotype

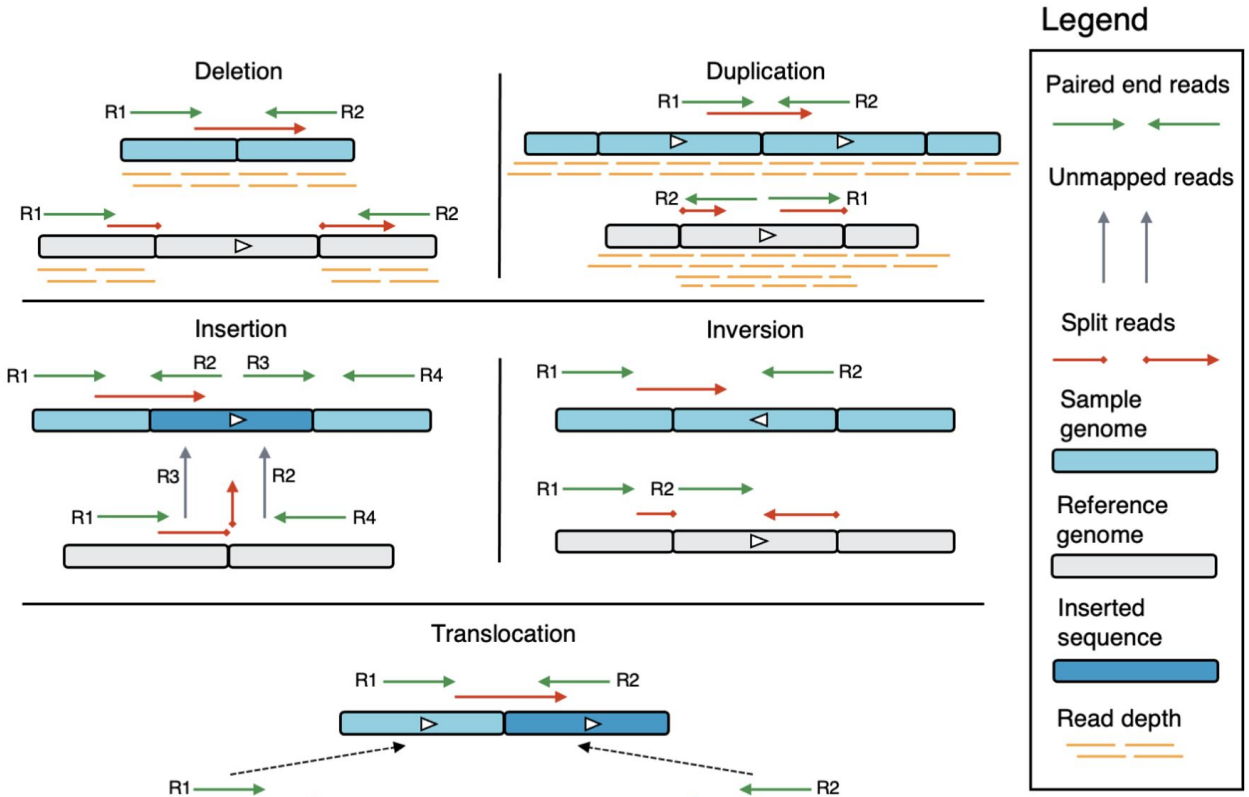


Illumina (“short reads”)



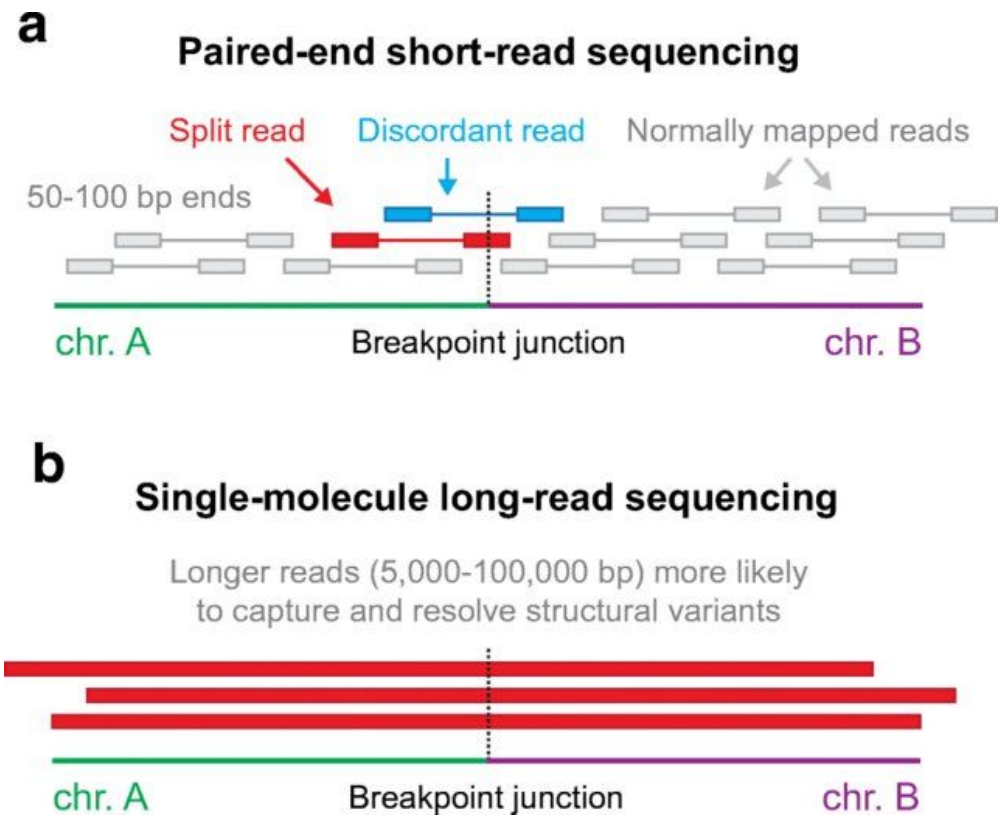
Hu et al. 2020

b,c) Short read evidence categories to find CNVs and SVs

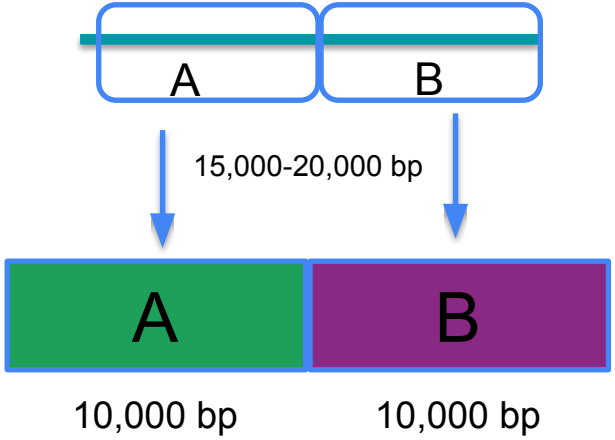


gatk.broadinstitute.org

c,d) Reads spanning breakpoints inform on junction and full haplotype

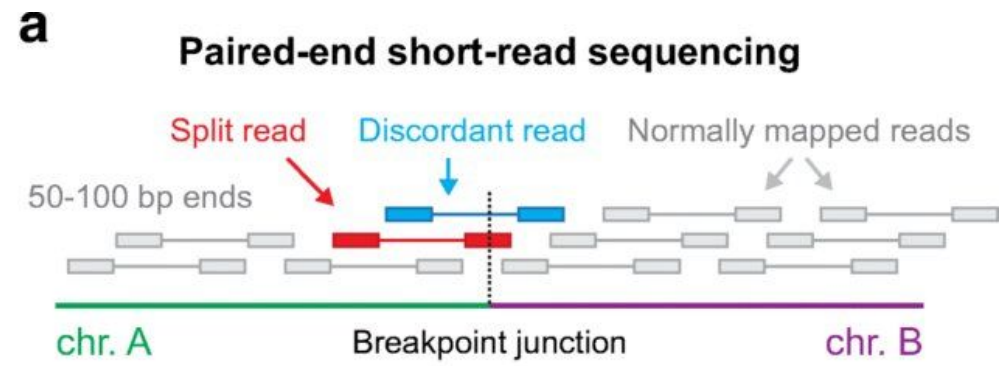


PacBio (“long reads”)

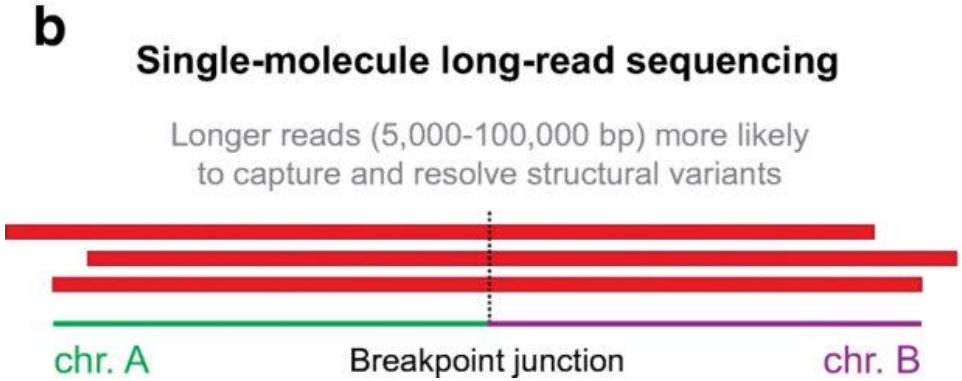


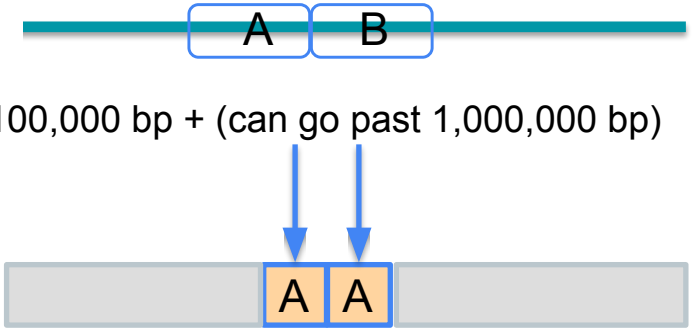
Hu et al. 2020

c,d) Reads spanning breakpoints inform on junction and full haplotype



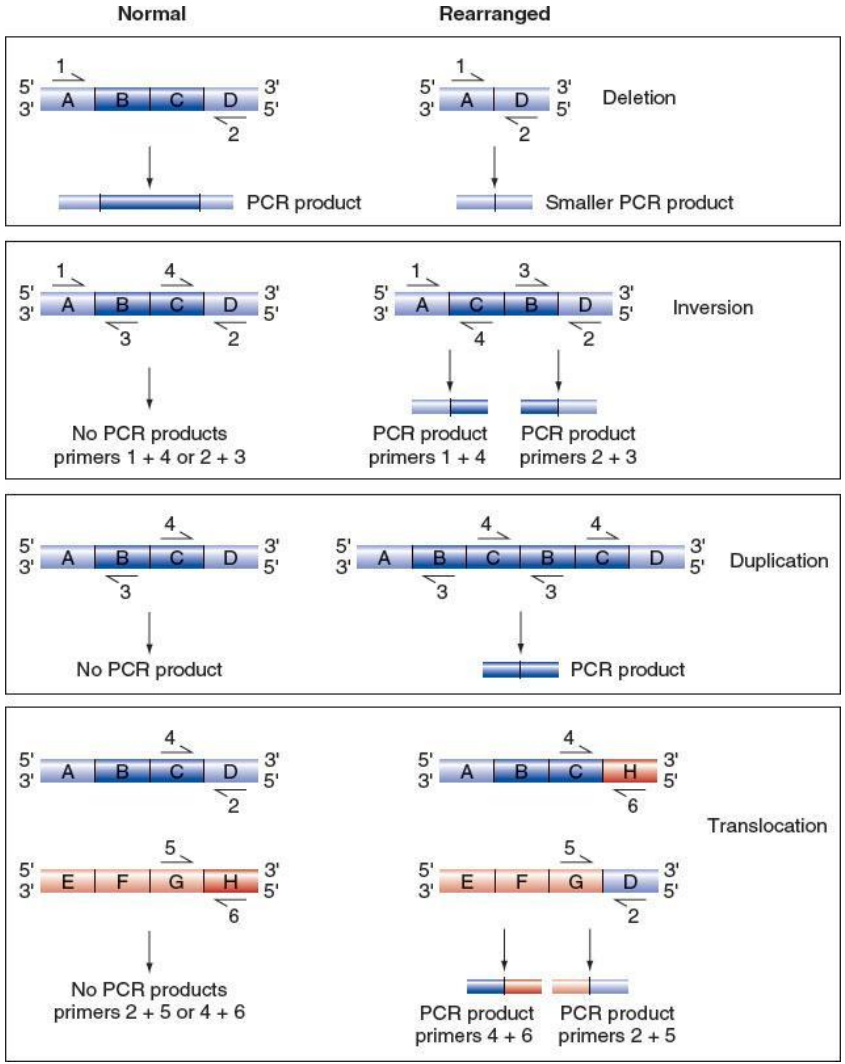
Nanopore (“ultra-long reads”)





Hu et al. 2020

PCR validation of chromosomal alterations



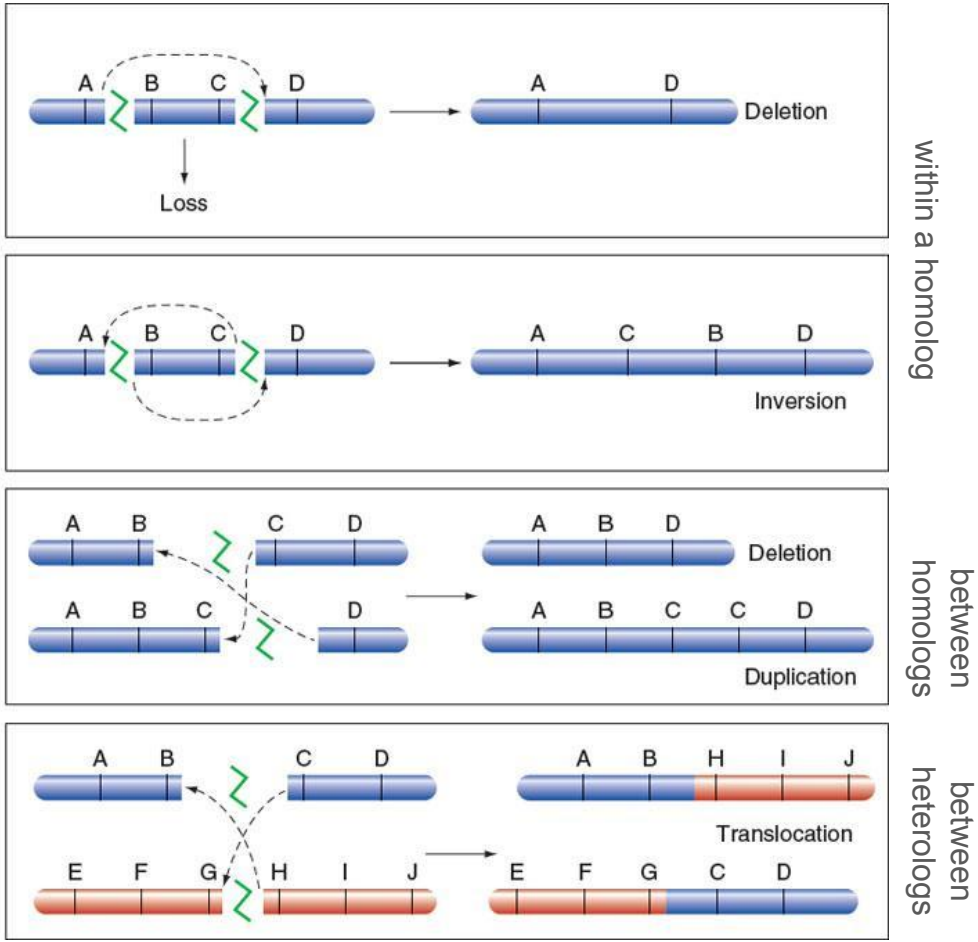
IV. Chromosomal rearrangement generation mechanisms: <u>Breakage and Repair</u>

Green ‘Z’s indicate breakpoints

Arrows indicate which pieces were repaired and joined together

Homologous chromosomes

Non-homologous chromosomes

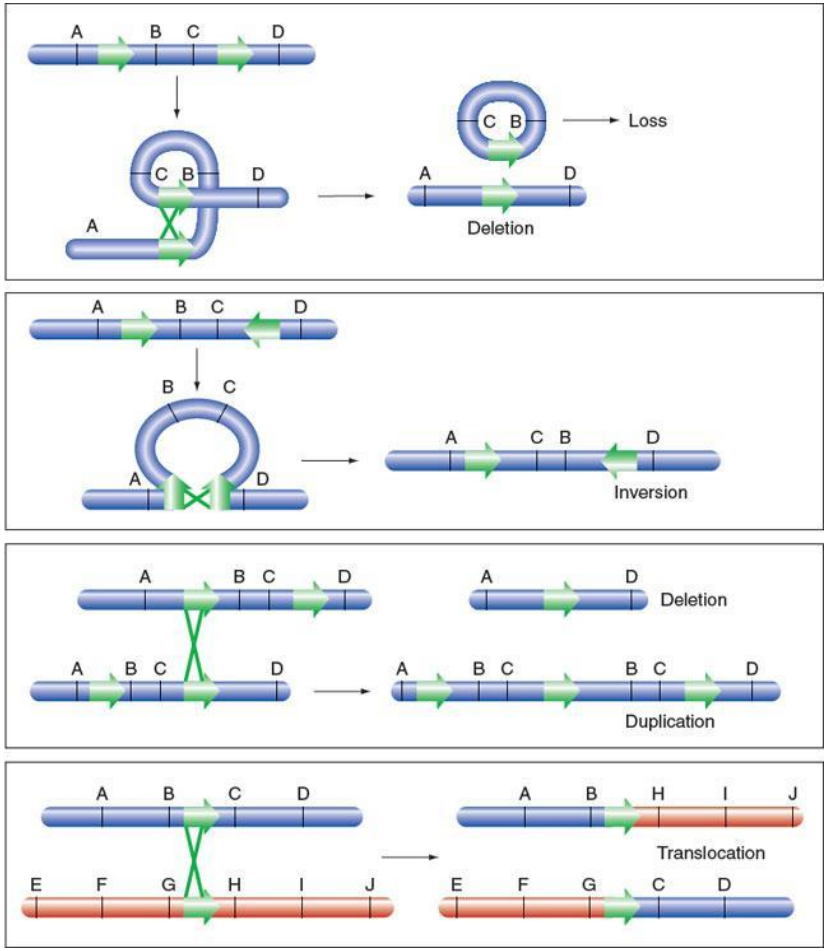


IV. Chromosomal rearrangement generation mechanisms: <u>Aberrant cross-over</u>

Green arrows indicate similar sequences that pair during meiotic crossover

Homologous chromosomes

Non-homologous chromosomes

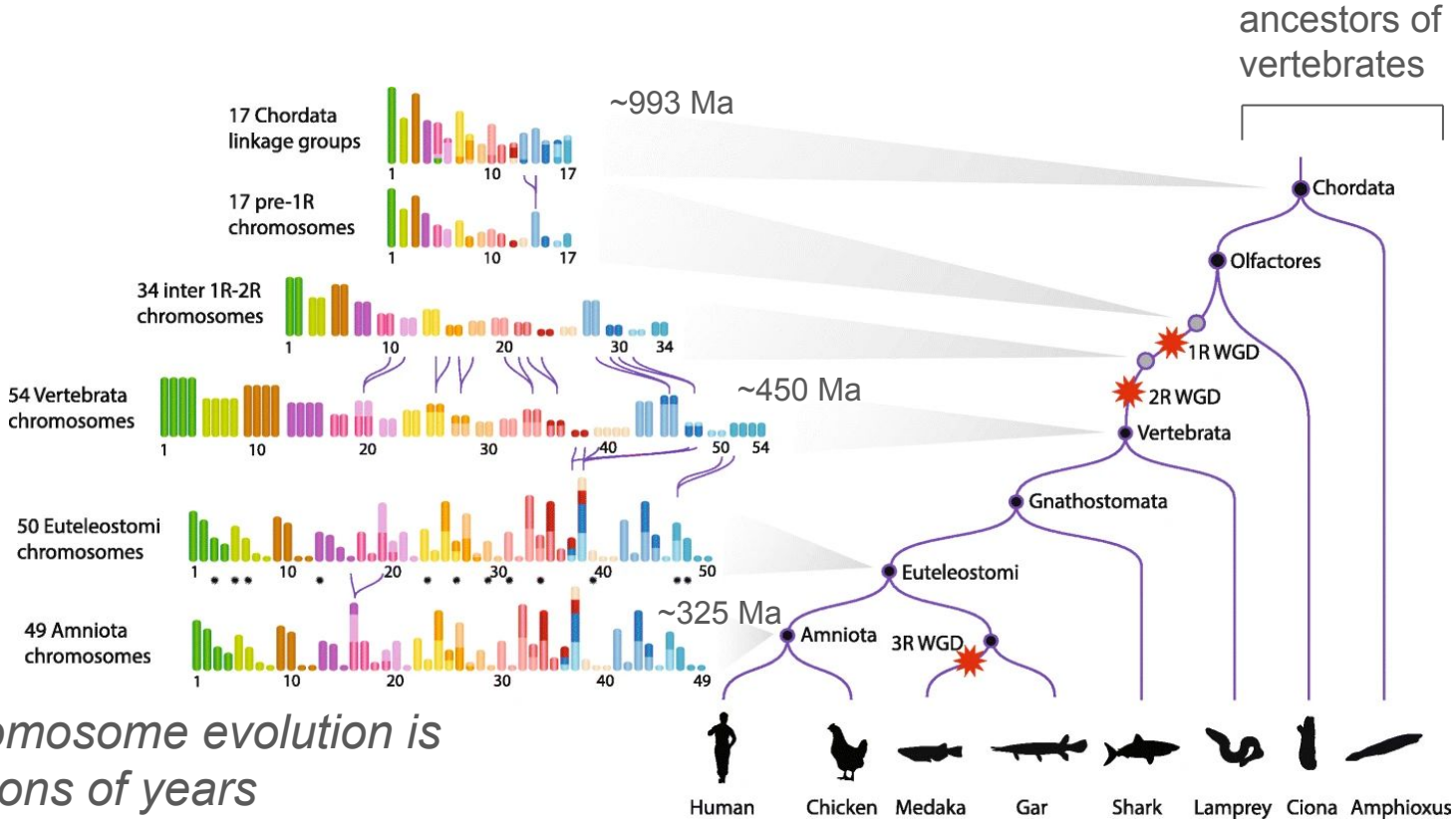


V. Evolutionary implications of chromosomal rearrangements

Reconstructed evolutionary history of karyotypes from Chordata to Amniota

WGD = whole genome duplication

Vertebrates = organisms with backbones



⇒ *Vertebrate chromosome evolution is dynamic over millions of years*